

Photutils Optimisation Parameters Explained

Foreground-object masking for barred-galaxy profile analysis

Purpose. The optimisation should find the minimum intervention that removes narrow foreground-object spikes from the bar-major profile while preserving genuine galaxy structure such as bars, rings, spiral arms, ansa, shoulders, and nuclear components.

Why These Parameters Matter

The Photutils masking pipeline works on a residual image: the science image minus a broad Gaussian-smoothed galaxy model. Compact positive residuals are treated as foreground-candidate sources. The optimiser then tests parameter combinations and scores them against two kinds of evidence: spike-positive galaxies, where the mask should cover narrow bar-profile spikes, and no-spike control galaxies, where the mask should not change a clean profile.

The parameters below control three linked decisions: what counts as a detection, which detections are kept, and how aggressively retained detections are expanded into masks.

Parameter Summary

Parameter	Current or proposed values	Main role
nsigmas	5.5, 5.0, 4.5, 4.0, 3.5	Detection threshold in residual-sigma units.
dilations	1, 2, 3 pixels	Expands retained masks to include source wings.
max-areas	150, 300, 500 pixels	Rejects large residual regions that are likely galaxy structure.
max-elongation	3, 4, 6	Rejects stretched detections such as arms, bars, streaks, or diffuse residuals.
smooth_sigma_pixels	15 px currently; test 10, 15, 20, 25	Sets the scale of the smooth galaxy model subtracted before detection.
npixels	8 px currently; secondary test values could include 5, 8, 12	Minimum connected pixels required for a detection.
central_exclusion	12 px currently; secondary test values could include 8, 12, 16, 20	Protects the nucleus and compact nuclear structures from being masked.

Detailed Parameter Effects

Parameter	If made more aggressive	If made more conservative	Optimisation signal
nsigmas	Lower nsigma detects fainter residuals and may catch weaker foreground objects.	Higher nsigma ignores fainter residuals and reduces false positives.	Choose the highest threshold that still gives high spike coverage.
dilations	Larger dilation captures PSF wings and nearby foreground-object light.	Smaller dilation preserves more local galaxy light around each detection.	Penalise large affected-profile fractions and control-galaxy damage.
max-areas	Larger max area keeps broader detections, which may catch big stars but risks galaxy features.	Smaller max area rejects broad residual patches but can miss large foreground objects.	Prefer the smallest area cap that does not reduce spike coverage.
max-elongation	Larger values keep more stretched detections.	Smaller values reject elongated residuals more strongly.	Use controls to check that spiral arms, bars, and streak-like galaxy residuals are not retained.
smooth_sigma_pixels	A larger smoothing scale leaves more medium-scale residual structure in the detection image.	A smaller smoothing scale can absorb compact sources into the model or create local subtraction artefacts.	Optimise by checking whether residual sources are enhanced without turning galaxy structure into detections.
npixels	Lower npixels accepts smaller connected detections and increases sensitivity to tiny objects or noise.	Higher npixels rejects small detections and suppresses noise.	Keep fixed initially; tune only if many real small contaminants are missed or noise detections are common.
central_exclusion	A smaller exclusion allows detections nearer the centre, increasing risk to the nucleus.	A larger exclusion protects more central light but may miss foreground objects projected near the centre.	Tune only after checking central cases visually; avoid optimising this from spike coverage alone.

How To Optimise

The existing optimiser already scores global Photutils parameter sets using spike galaxies and no-spike controls. It should be extended so `smooth_sigma_pixels` can vary in the main grid. The secondary parameters `npixels` and `central_exclusion` should usually remain fixed until the primary grid has identified a stable region of good solutions.

Tier	Parameters	Reason
Primary grid	<code>nsigma</code> s, <code>dilation</code> s, <code>max-areas</code> , <code>max-elongation</code> , <code>smooth_sigma_pixels</code>	These have the strongest effect on sensitivity, false positives, and profile damage.
Secondary checks	<code>npixels</code> , <code>central_exclusion</code>	These are important safeguards but are easier to overfit, especially with a small calibration set.
Final visual review	Best 3-5 scored parameter sets	The score narrows the field; visual reports catch scientific edge cases that simple metrics miss.

Scoring Logic

The score should reward spike coverage and penalise unnecessary masking. A good parameter set should cover bar-profile spike samples in spike-positive galaxies, while producing low masked-pixel fractions and low profile-shape changes in no-spike control galaxies.

Practical rule. Select the most conservative parameter set whose spike coverage is effectively complete and whose control galaxy damage remains low. If two settings perform similarly, prefer the one with the higher `nsigma`, smaller `dilation`, smaller `max area`, and lower control-profile damage.

Spike-Gated Versus Global Photutils

Spike-gated masking can tolerate lower detection thresholds because Photutils detections are only candidates: a source is applied to the science profile only if it intersects detected bar-major spike samples. Global Photutils lacks that profile gate, so it should normally use stricter settings. For that reason, an `nsigma` value around 3.5 may be acceptable in spike-gated mode but should be treated as aggressive for global masking.

Recommended Next Step

Run the current optimiser over the existing nsigmas, dilations, max-areas, and max-elongation grid, then add smooth_sigma_pixels to the grid and compare the ranked summaries. Keep npixels at 8 and central_exclusion at 12 px for the first pass. Only vary them if the visual reports show missed small sources, noise detections, or central masking problems.